Torque and voltage dq identities (Issue #68)

Sensitivity to PointPer and mesh size

1 Identity checks and setup

For two 5-phase SynRM models (40 slots and 60 slots), we test the *amplitude-invariant* Park identities against Ansys Maxwell 2D:

$$T_{dq} = \frac{m}{2}\, p\, \left[(\psi_{d1} i_{q1} - \psi_{q1} i_{d1}) + 3(\psi_{d3} i_{q3} - \psi_{q3} i_{d3})\right], \tag{1}$$

$$V_{q1} - R_s I_{q1} = \omega \psi_{d1}, \qquad V_{d1} - R_s I_{d1} = -\omega \psi_{q1}. \tag{2}$$

The third-harmonic voltage checks use subscript 3 and 3ω; $m = 5$ (phases), p is the pole-pair count, and $\omega = 2\pi f$. For each check, A and B denote the period means of the two compared quantities (the left and right sides for voltage, or T_{dq} and `Moving1.Torque` for torque). The relative error is

$$\varepsilon_{\text{rel}} = \frac{|A - B|}{|B|} \times 100\%. \tag{3}$$

Period means use `np.mean`, the discussion criterion is 5%.

The tests use $f = 50\,\text{Hz}$, $N_{\text{per}} = 1$, fixed currents $(i_{d1}, i_{q1}, i_{d3}, i_{q3}) = (0.8, 0.9, 0.2, -0.15)\,\text{A}$ and the OneLambda rotor (seed 42). The time step is $\Delta t = 1/[f(\text{PointPer} - 1)]$. The 3 mm reference mesh is compared with a 0.5 mm maximum element size applied to the *stator, rotor, air gap and coils*.

2 Sensitivity results

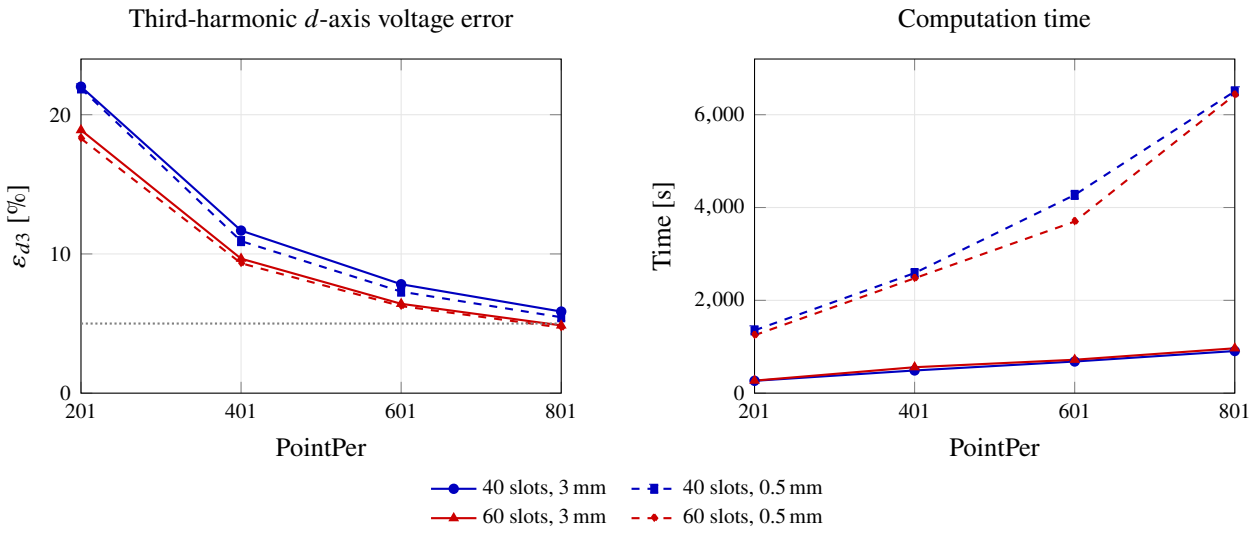


Figure 1: Mesh comparison for the common PointPer range (201–801). The dotted line marks the 5% error threshold.

PointPer matters more. At 3 mm, ε_{d3} drops from 22.02% to 5.86% (40 slots) and from 18.89% to 4.87% (60 slots) as PointPer increases from 201 to 801. All five errors on the 3 mm mesh first fall below 5% at PointPer 1001 (40 slots) or 801 (60 slots).

Finer meshing offers little improvement. At PointPer 801, refining from 3 to 0.5 mm reduces ε_{d3} only from 5.862% to 5.453% (40 slots) and from 4.866% to 4.712% (60 slots), but raises runtime by 7.2 times and 6.6 times.

Conclusion. Increasing PointPer is more effective than refining the mesh for these dq checks. When criterion is set to 5%, the values of PointPer are 801 for 40 slots and 1001 for 60 slots, respectively.

3 Complete sweep data

All 21 cases' raw data are listed below. P = PointPer; $h_{\max}$ = maximum mesh length [mm]; t = elapsed computation time [s]. ε_T is the torque-identity error; ε_{q1}, ε_{d1}, ε_{q3} and ε_{d3} are the first- and third-harmonic q-/d-axis voltage identity errors. All five errors are in percent.

Table 1: Tested cases (errors in %; bold rows: all five errors below 5%).

				Relative error [%]				
Slots	P	$h_{\max}$ [mm]	t [s]	ε_T	ε_{q1}	ε_{d1}	ε_{q3}	ε_{d3}
40	201	3	266	0.1452	0.2168	8.8363	0.9687	22.0154
40	401	3	489	0.3381	0.1070	4.7957	0.4139	11.6775
40	601	3	682	0.3864	0.0781	3.1646	0.3021	7.8227
40	801	3	908	0.4118	0.0593	2.3503	0.2384	5.8617
40	**1001**	**3**	**1981**	**0.4164**	**0.0479**	**1.8624**	**0.1937**	**4.6725**
40	**1201**	**3**	**1370**	**0.4206**	**0.0400**	**1.5430**	**0.1625**	**3.8838**
40	**2001**	**3**	**4169**	**0.4300**	**0.0241**	**0.9144**	**0.0980**	**2.3215**
60	201	3	270	0.2944	0.3223	9.1336	0.8636	18.8941
60	401	3	559	0.1740	0.1391	4.1977	0.4237	9.6603
60	601	3	720	0.1738	0.0889	2.7516	0.2925	6.4179
60	**801**	**3**	**967**	**0.2028**	**0.0737**	**2.1236**	**0.2492**	**4.8662**
60	**1201**	**3**	**2003**	**0.2221**	**0.0531**	**1.4476**	**0.1812**	**3.2774**
60	**2001**	**3**	**76474**	**0.2313**	**0.0322**	**0.8652**	**0.1090**	**1.9687**
40	201	0.5	1353	0.2507	0.2074	8.8112	0.9326	21.8884
40	401	0.5	2588	0.0076	0.1114	4.3557	0.4760	10.9223
40	601	0.5	4272	0.0171	0.0751	2.9180	0.3214	7.2923
40	801	0.5	6504	0.0270	0.0571	2.1867	0.2425	5.4529
60	201	0.5	1248	0.1930	0.3048	9.3010	0.8571	18.2915
60	401	0.5	2476	0.1574	0.1594	4.4569	0.5135	9.3301
60	601	0.5	3698	0.1231	0.1067	2.9596	0.3531	6.2536
60	**801**	**0.5**	**6428**	**0.1031**	**0.0810**	**2.2027**	**0.2642**	**4.7123**